

Practicum No.5**Date:13/2/2026****Customer Segmentation Analysis for a CSE/AIML/DSCS/DS Start-up****I. Practical Significance**

This practicum enables students to systematically identify and analyze distinct customer segments for a proposed CSE/AIML/DSCS/DS product or service. Understanding segmentation helps a start-up define user personas, analyze behavioural traits, map digital adoption levels, and align product features with market expectations. By performing segmentation, students gain insights into who their real customers are, what needs they prioritize, and how they differ in motivations, willingness to adopt, and usage patterns. This exercise supports the development of early-stage strategies for SaaS tools, AI/ML models, cybersecurity services, cloud-native platforms, intelligent automation tools, and data-driven applications.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a pitch deck, project proposals for small/medium scale startups logically and scientifically.
- a) **Interpret** target customer clusters for SaaS applications, AI-powered automation tools, and data-driven service platforms. (CDO)
- b) **Evaluate** commercial viability of CSE products such as AI models, software applications, analytics platforms, and cloud-native solutions. (PDO)
- c) **Practice** professional ethics in data collection, system design, and user interaction within digital ecosystems. (ADO)

III. Relevant CO(s)

- **CO2. Validate** feasible business ideas using alternative entrepreneurial methods.

IV. Practical Outcome (PrO)

- **PrO5 –Conduct** a customer segmentation analysis for **the identified** CSE, AIML, DSCS & DS start-up.

V. Related ADO(s) (keep as such)

- a) **Practice** professional ethics.
- b) **Function** effectively as a team member/leader.
- c) **Practice** integrity and honesty.

VI. Minimum Underpinning Theory

Customer segmentation is the process of grouping users based on shared characteristics to better understand their needs, preferences, and adoption trends.

Common segmentation models used in technology ventures include:

- Demographic Segmentation: Age, education, role, domain.
- Behavioural Segmentation: Digital tool usage, purchase behaviour, engagement levels.
- Psychographic Segmentation: Attitudes, motivations, adoption readiness.
- Technographic Segmentation: Familiarity with emerging technologies like AI, ML, cloud, DevOps tools, cybersecurity systems.

Segmentation helps:

- Identify early adopters and high-value users.
- Prioritize features for different customer profiles.
- Support pricing strategy and go-to-market planning.

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VIII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For research and documentation	1 per team
2	Internet access	For identifying trends and problems	As required
3	Survey/Interview Data	Collected earlier or created	1 set
4	Spreadsheet	Excel	1 set

IX. Safety Precautions

- Avoid** collecting unnecessary personal or sensitive data.
- Maintain** confidentiality of respondent information.
- Use** the collected data only for academic/educational purposes.
- Obtain** consent before collecting responses.
- Ensure** data anonymity when documenting results.
- Follow** ethical norms of transparency and voluntary participation.

X. Procedure

- Select** a CSE/AIML/DSCS/DS product or service.
- Identify** the purpose of segmentation.
- Choose** segmentation criteria (demographic/behavioural/technographic/psychographic).
- Collect** user data via surveys, interviews, or secondary sources.
- Evaluate** data to identify natural clusters.
- Group** customers into meaningful segments.
- Prepare** detailed segment profiles.
- Present** segment insights using tables, charts, or matrices.

XI. Actual Resources Used *(To be filled by the students)*

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	HP	Used to analyze collected data and prepare the report	1	Personal device
2.	Internet Connection	WiFi	Used for research and accessing survey responses	1	Required for analysis
3.	Survey Data	Google Forms	Responses collected from students about career tools	1 set	Used for segmentation
4.	Spreadsheet Software	Microsoft Excel	Used to organize and group customer segments	1	For analysis

XII. Actual Procedure Followed

- First, we selected **Digital Attendance and Productivity Tracker** as the startup product idea for the customer segmentation analysis.
- Then we defined the **purpose of segmentation**, which was to understand different types of users who may benefit from a digital attendance and productivity tracking system.

3. After that, we selected **segmentation criteria such as demographic (role and education), behavioural (attendance management habits), psychographic (motivation to improve productivity), and technographic (familiarity with digital tools).**
4. Basic user data was collected from **survey responses obtained in the previous practicum and simple assumptions about student and teacher behaviour.**
5. The collected data was organized and analyzed using **spreadsheet tools such as Microsoft Excel.**
6. Based on the analysis, users were grouped into segments such as **students who need attendance tracking, teachers who manage attendance records, and tech-savvy users interested in productivity analytics.**
7. Finally, the characteristics, needs, and expectations of each segment were analyzed to understand how the product can meet their requirements.

XIII. Observations and Data

S. No.	Customer Attribute / Segmentation Variable	Data/ Remarks
1	User Role	Respondents included both students and teachers
2	Attendance Management Method	Many users rely on manual attendance or simple digital systems
3	Technology Usage	Most students are comfortable using mobile apps and digital platforms
4	Productivity Monitoring	Teachers expressed interest in tools that can track student performance
5	Motivation for Using System	Users want easier attendance management and productivity insights

XIV. Results/Observations

1. The segmentation analysis identified **three main user segments: students, teachers, and tech-savvy academic users** interested in productivity monitoring.
2. Students mainly need **easy attendance tracking and reminders**, while teachers require **tools to record attendance and generate reports.**
3. Tech-savvy users prefer **advanced features such as productivity analytics and automated reports.**

XV. Interpretation of Results

1. The segmentation analysis shows that **teachers and students are the primary users** of the Digital Attendance and Productivity Tracker.
2. Teachers are more likely to use the system for **attendance management and reporting**, while students benefit from **tracking their academic participation and productivity.**
3. Understanding these segments helps in designing features that meet the needs of different users.

XVI. Conclusions

1. From this segmentation analysis, it can be concluded that students and teachers form the main customer segments for the Digital Attendance and Productivity Tracker.
2. Teachers represent the primary decision makers, as they manage attendance records and academic monitoring.

- Students represent regular users, as they interact with the system frequently to track attendance and productivity.
- By focusing on these segments, the startup can design features that improve attendance management, productivity tracking, and academic performance monitoring.

XVII. Suggested Practicum Related Questions

- Evaluate the relevance of the chosen segmentation variables for a computing-oriented product.
- Evaluate how behavioural and technographic segmentation improves precision in product positioning.
- Examine possible overlaps between segments and propose methods to refine classifications.
- Assess the implications of identified segments for pricing, feature prioritization, and early adopter engagement.
- Interpret how segmentation supports the development of MVP and go-to-market strategies.

XVIII References / Suggestions for further reading (keep as such)

- Venture Meets Mission: Aligning People, Purpose, and Profit to Innovate and Transform Society, Arun Gupta, Gerard George, Thomas J. Fewer, Stanford University Press, Stanford, California, 2024, ISBN: 978-1503636283
- So You Want to Start a Business: Eight Steps to Startup Success, John B. Vinturella, Business Expert Press, New York, USA, 2024, ISBN: 978-1637425855
- Business Planning, Franklin A. Gevurtz, Foundation Press (West Academic), St. Paul, Minnesota, 2024, ISBN: 978-1685612764

XIX Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Data collection and preparation	3	
2	Selection of segmentation parameters	4	
3	Analytical reasoning	4	
4	Team collaboration	3	
5	Ethical handling of customer data	4	
Product-related Skills: 12 Marks - 40%			
1	Clarity of segmentation model	3	
2	Accuracy of segment interpretation	3	
3	Structure and quality of documentation	3	
4	Timely submission	3	
Total		30	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	Neavis Rishva. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained	Name and Designation of the Faculty Member	Signature with Date
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Process-Related skills (18)	Product-Related skills (12)	Total (30)		

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